

Vibe Coding for Data Visualization

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1. Workshop Objectives

- Understand the vibe coding workflow and how it transforms traditional data visualization.
- Guide AI agents to style chart anatomy (axes, labels, ticks, legends) using qualitative feedback.
- Select and orchestrate the right chart types (line, bar, pie, box, scatter plots) using natural language directives.
- Prompt the Google Antigravity Agent using Gemma-4-31B-QAT and NVIDIA Nemotron-3 to generate matplotlib, seaborn, and plotly code.
- Master the iterative feedback loop to debug, refine, and polish complex interactive dashboards.

2. Chart Anatomy & Type Selection

Core Chart Anatomy

Axes	The horizontal and vertical reference lines (x/y axes).
Labels	Explanatory text identifying axes and data categories.
Ticks	Grid markers and numeric values showing scaling.
Legends	Key explaining visual markers, colors, or line styles.

Chart Type Selection

- **Line Chart:** Continuous data changes over time / trends.
- **Bar Chart:** Comparing sizes of discrete categorical items.
- **Pie Chart:** Part-to-whole proportions (small groups).
- **Box Plot:** Statistical distribution and outliers.
- **Scatter Plot:** Relationships between continuous variables.

3. Stack & Setup

Environment & Tools

- IDE: Google Antigravity IDE (VS Code based)
- Language: Python 3.9+
- Core Packages: matplotlib, seaborn, plotly, pandas
- AI Assistance: Google Antigravity Agent

AI Models

Gemma-4-31B-QAT

Optimized for rapid drafting and standard plotting library code.

NVIDIA Nemotron-3

Superb for complex logical reasoning, custom layouts, and debugging.

4. Workshop Agenda (3 Hours)

Time	Module	Key Topics	Deliverables
00:00-00:30	1. Intro & Setup	Vibe coding concept, anatomy, env config	Environment verified
00:30-01:10	2. Iteration Loop	Draft zero, aesthetic iteration, fine tuning	Labs 1 & 2
01:10-01:20	3. Break	-	-
01:20-02:10	4. Dashboards	Plotly, interactive layout, model comparison	Lab 3
02:10-02:45	5. Challenges	Debugging, style mimicry	Lab 4
02:45-03:00	6. Wrap-Up	Showcase, best practices, Q&A	Interactive feedback

5. Lab 1: Marketing Channel Distribution

Goal

Visualize how prospective international students first hear about the GMBA program.

Visual Principle

Sorting & Label Clutter: Always sort categorical charts by frequency (descending). Rotate labels by 45° to prevent overlaps.

Vibe Coding Tip: Tell the agent to clean white spaces, aggregate using value counts, and apply a specific rotation factor to the x-tick labels.

5. Lab 2: Importance vs Sufficiency Gap

Goal

Compare perceived importance of homepage info (Q3) vs current sufficiency (Q4).

Visual Principle

Grouped Charts: Side-by-side bars highlight discrepancies. Map Likert scales (text) to numeric integers (1-5) before averaging.

Vibe Coding Tip: Instruct the agent to map textual Likert scales to numeric integers using pandas mapping.

5. Lab 3: Usability Barriers Heatmap

Goal

Chart specific navigation and usability barriers (Q6) students face.

Visual Principle

Matrix Data: Melt wide tables into long format. Use sequential color gradients (heatmaps) to draw attention to pain points.

Vibe Coding Tip: Prompt the agent to use pandas `melt` function to reshape the columns before plotting.

5. Lab 4: Satisfaction Dashboard

Goal

Assemble a unified HTML dashboard: Overall satisfaction (Q10) + Preferred formats (Q9).

Visual Principle

Hierarchy: KPIs top-left (F-pattern). Group details to right/bottom. Ensure color consistency.

Vibe Coding Tip: Direct the agent to generate a combined Plotly grid layout or a simple HTML/CSS template.

6. Prompting Tips for Data Viz AI

Structure: Intent → Context → Aesthetic Constraints

```
"Act as a clean, modern dashboard designer. Using the loaded sales dataframe,
generate a matplotlib scatter plot showing marketing spend versus revenue.
- Group data points by product category using distinct, soft-toned colors.
- Format the ticks to show dollar signs.
- Keep the style minimal with a light gray grid and no outer borders."
```